

FORT MADISON, IA, USA

WMO: 725483

Lat: 40.659N

Lon: 91.327W

Elev: 221

StdP: 98.70

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04930

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.7	-15.0	-22.5	0.5	-16.9	-20.0	0.6	-14.1	12.0	-4.5	10.9	-4.9	3.3	310	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	33.9	25.6	32.5	25.2	31.3	24.3	27.4	32.1	26.3	30.8	25.4	29.9	3.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.2	22.2	30.4	25.0	20.6	29.5	23.8	19.2	28.4	88.1	32.5	83.3	31.0	78.6	30.4	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.5	7.6	DB	-21.9	35.8	3.7	1.8	-24.6	37.1	-26.7	38.2	-28.8	39.2	-31.4	40.5	(4)
(5)			WB	-21.4	28.6	4.2	0.7	-24.5	29.0	-26.9	29.4	-29.3	29.8	-32.3	30.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	-3.3	-1.6	5.5	11.8	18.0	22.7	24.6	23.8	20.0	13.2	5.6	-0.1	(6)	
	DBStd	11.13	6.75	6.34	6.52	5.64	4.76	3.41	3.24	3.08	4.39	5.35	5.61	5.86	(7)	
	HDD10.0	1452	414	325	171	44	2	0	0	0	0	28	154	314	(8)	
	HDD18.3	3068	672	557	401	206	66	6	1	1	32	174	384	570	(9)	
	CDD10.0	2096	1	2	31	98	251	382	454	427	301	128	21	2	(10)	
	CDD18.3	670	0	0	2	10	57	137	197	169	82	16	1	0	(11)	
	CDH23.3	6202	0	0	14	96	479	1227	1989	1531	741	124	2	0	(12)	
(13)	CDH26.7	2283	0	0	1	17	142	444	807	583	260	30	0	0	(13)	
(14)	Wind	WSAvg	3.3	4.0	4.0	4.2	4.2	3.5	2.9	2.3	2.0	2.4	3.1	3.8	3.7	(14)
(15) Precipitation	PrecAvg	952	45	44	66	97	122	117	103	99	77	71	64	45	(15)	
	PrecMax	1310	81	125	138	155	293	261	297	190	153	185	174	111	(16)	
	PrecMin	700	13	6	18	57	20	19	38	30	17	28	0	9	(17)	
	PrecStd	178	21	31	38	27	73	66	68	51	46	44	42	27	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.1	17.1	25.9	29.1	32.6	34.0	36.3	36.1	34.0	30.1	22.5	17.2	(19)	
		MCWB	11.1	11.3	17.7	19.7	23.3	25.0	26.2	24.9	24.0	21.9	14.5	13.8	(20)	
	2%	DB	11.2	12.7	22.1	26.3	30.0	32.5	34.0	33.6	32.1	27.0	18.9	12.8	(21)	
		MCWB	8.3	8.6	15.5	17.2	22.2	24.6	26.3	25.5	23.4	18.9	13.9	9.9	(22)	
	5%	DB	7.8	9.8	18.0	23.6	27.8	31.2	32.5	32.1	29.9	23.9	17.0	10.5	(23)	
		MCWB	5.2	6.8	13.4	16.0	20.1	23.8	26.0	25.3	22.8	17.6	12.3	8.5	(24)	
	10%	DB	5.2	7.2	15.2	21.3	26.2	29.1	31.2	30.0	27.6	22.1	13.8	7.7	(25)	
		MCWB	2.7	4.7	10.9	14.7	18.9	23.1	25.0	24.0	21.0	16.8	9.9	5.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	12.1	18.5	20.8	25.1	27.6	29.2	28.4	26.2	22.9	16.6	14.9	(27)	
		MCDB	14.0	15.7	24.0	27.2	30.7	32.0	33.5	33.7	30.7	28.3	19.2	17.2	(28)	
	2%	WB	8.0	9.3	16.4	18.7	23.1	26.0	28.1	26.8	24.9	20.9	14.4	11.0	(29)	
		MCDB	10.2	12.7	20.0	24.1	28.3	30.5	32.4	31.8	29.8	24.1	17.9	12.3	(30)	
	5%	WB	5.4	6.9	14.4	17.3	21.8	25.0	26.7	25.7	23.7	19.0	12.3	8.0	(31)	
		MCDB	8.4	9.2	18.0	21.8	26.6	29.5	31.0	30.4	27.9	22.8	15.9	9.6	(32)	
	10%	WB	3.2	4.6	11.5	15.6	20.3	23.9	25.6	24.7	22.4	17.0	10.2	5.7	(33)	
		MCDB	5.1	7.1	15.0	20.5	24.4	28.1	30.0	28.8	26.4	20.9	13.6	7.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	8.4	9.7	10.8	10.6	10.1	10.2	10.5	11.9	10.6	9.0	7.6	(35)	
		MCDBR	11.1	11.9	12.9	14.1	12.5	11.5	10.7	12.1	12.6	12.9	12.0	10.5	(36)	
	5% WB	MCWBR	8.9	8.9	8.3	7.6	6.3	5.9	5.8	6.7	6.1	7.1	8.4	8.7	(37)	
		MCWBR	10.6	10.9	11.9	12.3	11.1	10.2	10.1	10.5	10.8	10.4	11.1	10.2	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.290	0.308	0.339	0.389	0.422	0.425	0.443	0.433	0.395	0.349	0.302	0.289	(40)	
		taud	2.453	2.400	2.404	2.266	2.236	2.284	2.207	2.255	2.312	2.443	2.515	2.473	(41)	
	Ebn at Noon	Ebn at Noon	875	907	913	883	858	854	833	832	840	848	856	848	(42)	
		Edh at Noon	77	95	106	131	138	132	141	131	115	91	73	70	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.84	2.79	3.71	4.77	5.57	6.24	6.23	5.48	4.70	3.14	2.11	1.57	(44)	
	RadStd	RadStd	0.22	0.24	0.26	0.38	0.40	0.58	0.40	0.30	0.34	0.35	0.17	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page